

Part 01 :

Q1 : Explain with code example how class and struct behave differently

Q2 : Explain the difference between public and private access modifiers with an example.

Q3 : Describe the steps to create and use a class library in Visual Studio.

Q4 : What is a class library? Why do we use class libraries?

ROUTE

Part 02 :

Movie Ticket Booking System

User Story: You're building a simple Movie Ticket Booking System for a cinema. The system manages ticket types, seat locations, pricing, and payments. Build it as a Console Application that reads data from the user and prints the booking summary.

what you need to build :

1. Each ticket has a type that can only be one of: Standard, VIP, or IMAX. How would you represent this?
2. You need a type to represent a seat location (Row as a char like 'A', 'B', and Number as an int). Should this be a class or a struct? Create it.
3. Create a Ticket class with:
 - a. MovieName (public),
 - b. Type (public)
 - c. Seat (public)
 - d. Price (private).

Sometimes a ticket is created with all info, sometimes with just the movie name (default type Standard, seat A1, price 50). Handle both without repeating initialization logic.

4. Add three methods to the Ticket class:

- a. CalcTotal() — receives a taxPercent (double), calculates the total after tax and returns it. The original price must stay unchanged.
- b. ApplyDiscount() — receives a discountAmount (double) . If discount is valid (> 0 and $\leq \text{Price}$), deducts it from Price and sets discountAmount to 0 (consumed). Otherwise, the discount stays unchanged.
- c. PrintTicket() — prints the full ticket info.

Create a Console Application. Read the ticket data from the user, then print the following output:

Expected Output:

```
Enter Movie Name: Inception
Enter Ticket Type (0 = Standard , 1 = VIP , 2 =
IMAX ): 2
Enter Seat Row (A, B, C...): B
Enter Seat Number: 5
Enter Price: 200
Enter Discount Amount: 30

===== Ticket Info =====
Movie      : Inception
Type       : IMAX
Seat       : B5
Price      : 200.00
Total (14% tax) : 228.00

===== After Discount =====
Discount Before : 30.00
Discount After  : 0.00
Movie          : Inception
Type           : IMAX
```